

HAKIM AREZKI

HEAD OF IT AI PLATFORM — ENTERPRISE GENAI, LLM & AGENTIC PLATFORM

AI and Data Platform leader with 12+ years of experience (15+ including doctoral R&D) designing, industrialising and operating enterprise AI and Data platforms in regulated environments, including banking (BIL, BPCE) and industry (TotalEnergies). PhD, CentraleSupélec.

Runs AI capabilities as a shared enterprise platform: multi-LLM orchestration and model serving, AI agents, RAG and Document AI, reusable components (agents, prompts, APIs) offered to business domains in self-service within a secure and governed framework.

Treats the AI Platform as a product, with ownership of vision, roadmap, operating model and KPIs covering adoption, performance, reliability and cost — including FinOps control of LLM and API consumption — and AI lifecycle industrialisation through MLOps and LLMOps: deployment, versioning, CI/CD, evaluation, monitoring and continuous improvement.

AI governance and risk by design: access policies, model selection and approval, bias, hallucination and explainability controls, human validation, auditability, IAM and security of AI usage, GDPR and DORA — alignment with Data and Digital Services governance bodies. Strong data platform foundation on Snowflake feeding AI use cases with governed, trusted data.

BANKING & REGULATED ENVIRONMENTS

- **BIL Banque Internationale à Luxembourg (2026 - Present):** enterprise Data & AI platform, IBM Cognos to Snowflake / Power BI migration, RBAC and Row-Level Security, GDPR, DORA.
- **BPCE Banking & Insurance (2017 - 2018):** AML / CTF, sanctions and embargo screening, fraud detection, enterprise data catalog and lineage (Apache Atlas), GDPR compliance of financial data processing.
- **Cross-cutting:** regulatory reporting, auditability and data traceability, risk KPIs, IT governance and steering committees.

CORE SKILLS

Enterprise AI Platform (GenAI, LLMs, agents):

AI platform vision, roadmap, operating model and KPIs (usage, performance, adoption, cost), AI capabilities delivered as products, LLM platforms and APIs, model serving, multi-LLM orchestration with routing, caching and fallback, AI agents, tool calling and prompt frameworks, RAG, hybrid search and Document AI, AI marketplace and catalog of reusable components (agents, prompts, APIs, templates), self-service adoption and developer experience, MLOps and LLMOps (deployment, versioning, CI/CD, evaluation, monitoring, observability), FinOps control of LLM and API consumption, AI governance (access policies, model selection and approval, bias, hallucination and explainability controls, human-in-the-loop, auditability), security and IAM for AI usage, integration with Snowflake, enterprise applications and APIs

Data Platform Product Management:

Data Platform IT vision, roadmap and target operating model, Product-driven and federated model (core platform team + value stream teams), Platform KPIs: adoption, reliability, time-to-data, cost, Backlog definition and prioritisation of platform capabilities, Platform budget planning and consumption monitoring

Platform Run & Operations:

Service levels (SLA / SLO), availability and performance management, Monitoring, alerting, incident management and lifecycle management of platform services, Capacity planning and scalability, FinOps: Snowflake warehouse sizing, resource monitors, credit consumption tracking, query optimisation, cost allocation per business domain

Snowflake, Data Engineering & Data Management:

Enterprise data platform architecture on Snowflake, Batch and streaming ingestion frameworks, Data modelling and trusted / curated data layers, Industrialisation of transformation pipelines with dbt and Spark (Databricks, Kafka, Airflow), MDM, data quality, reconciliation and data lineage

BI, Power BI Cloud & Self-Service:

Power BI Cloud platform administration: workspaces, capacities, deployment pipelines, gateways, dataset refresh strategy, Semantic models, Power Query transformations, DAX measures and business KPIs, Row-Level Security and governed access, Migration of legacy BI estates to Power BI, Self-service analytics enablement, BI governance and dataset certification

Data Product & Governance:

Data product model: ownership, lifecycle and data contracts, Data catalog, data dictionary, lineage and marketplace (Apache Atlas, enterprise metadata), Alignment with the Data Office on governance, compliance and quality

Security, Risk & Compliance:

IAM / SSO, RBAC and Row-Level Security, Dynamic data masking and encryption, GDPR, DORA, auditability and traceability, Collaboration with Cybersecurity and Data Protection, AML / CTF, sanctions and embargo screening, financial crime and risk data governance

Leadership, Delivery & Governance:

Leadership of multi-disciplinary platform teams across Data, Engineering, Architecture, Cloud, DevOps and Business (teams of ~20 people), Vendor and external partner management, Architecture governance, ADRs and technical standards, Executive and C-level communication, Agile delivery: Scrum, Kanban, Jira, Confluence, Miro

TECHNOLOGY ENVIRONMENT

Data Platform & Cloud: Snowflake, Databricks, dbt, Azure, AWS

BI & Analytics: Power BI (Cloud, semantic models, Power Query, DAX, RLS, workspaces, deployment pipelines), IBM Cognos (migration source), Grafana

Data Engineering: Python, SQL, Spark, PySpark, Kafka, Airflow

Data Management & Governance: MDM, data quality, data lineage, data catalog, Apache Atlas, RBAC, RLS, masking, GDPR, DORA

AI & Generative AI: Azure OpenAI and Azure AI Foundry, OpenAI, Claude, Microsoft Copilot and Copilot Studio, Mistral, LLaMA, Hugging Face, RAG, AI agents, Model Context Protocol

AI & ML Frameworks: LangChain, LlamaIndex, Semantic Kernel, MLflow, Dataiku, PyTorch, scikit-learn, RAGAS

MLOps & Observability: MLflow, OpenTelemetry, Prometheus, Grafana, Langfuse, Arize

Infrastructure & DevOps: Docker, Kubernetes, Terraform, GitHub Actions, Azure DevOps, GitLab CI/CD

Databases & Integration: Snowflake, PostgreSQL, MongoDB, Redis, REST APIs, event-driven architecture

LANGUAGES

- French: bilingual / native-level proficiency
- English: fluent professional proficiency

LEAD DATA & AI PLATFORM: BIL, Banque Internationale à Luxembourg

01/2026 - Present

Modernisation of the bank's enterprise decision-support landscape: migration of a legacy IBM Cognos BI and Dataiku estate towards a modern Snowflake and Power BI data platform, together with the design of GENIA, the secure enterprise Data & AI platform supporting analytics, Machine Learning and Generative AI use cases within a regulated banking environment.

KEY ACHIEVEMENTS

Enterprise AI platform (GENIA): build, run and adoption

- Designed and industrialised the bank's enterprise AI platform, delivering GenAI, LLM and agent capabilities as shared, governed products to business domains, on Dataiku, Snowflake and enterprise data services.
- Designed a multi-LLM Mesh integrating Azure OpenAI, Claude and open-source models with intelligent routing, caching, fallback and latency control, and FinOps control of model and API consumption (usage, unit cost, p50/p95 latency, error rate).
- Built AI agents and RAG / Document AI pipelines consuming governed enterprise and customer data, with tool calling, state management, guardrails and human validation on sensitive actions.
- Integrated data and AI services through APIs and event-driven architectures using Kafka and Redis Streams.
- Implemented MLOps / LLMOps practices covering model and prompt lifecycle, versioning, CI/CD, evaluation gates before go-live, monitoring, traceability and continuous improvement.
- Built a catalog of reusable components — agents, shared system prompts, templates and APIs — to enable self-service adoption by business domains and improve developer and user experience.
- Contributed to the AI governance framework in line with Data & Digital Services governance: access policies, model selection and approval, usage rules, risk management (bias, hallucination, explainability), auditability of AI outputs, GDPR and DORA.
- Defined platform KPIs and steering indicators (usage, adoption, performance, reliability, cost) and reported them to Data, IT and business stakeholders.

Cognos and dataiku to Snowflake & Power BI migration and modernisation

- Led the modernisation of the legacy IBM Cognos decision-support platform towards a modern, secure and scalable Data & BI architecture based on Snowflake and Power BI.
- Ran the full inventory and dependency analysis of the existing Cognos estate — reports, packages, data models, source systems and embedded business rules — to map dependencies, rationalise the reporting portfolio and define the migration strategy and sequencing.
- Defined the target architecture and the progressive migration of data and processing to Snowflake, including the redesign of data models, transformation pipelines and consumption layers along secure and scalable patterns.
- Rebuilt Cognos reports and dashboards in Power BI: design of semantic models, Power Query transformations, DAX measures and business KPIs, and access management through Row-Level Security.
- Established a reconciliation framework between Cognos and Power BI outputs to guarantee figure-for-figure consistency, complemented by performance testing and data quality controls before each go-live.
- Drove user enablement and change management, and implemented BI governance covering ownership, modelling and naming standards, dataset certification and lifecycle management — securing the reliability, maintainability and adoption of the new platform.

Enterprise Data Platform (Snowflake)

- Contributed to the Data Platform vision and target architecture, translating business requirements into reusable platform capabilities and data products.
- Designed and evolved Snowflake-based data capabilities supporting analytics, reporting, Machine Learning and AI use cases.
- Worked with Data Engineering teams on data ingestion, transformation, data modelling and integration pipelines feeding the enterprise data platform.
- Defined trusted datasets and data products powering Power BI dashboards, business KPIs and self-service analytics.
- Contributed to platform cost control and FinOps practices: warehouse sizing, resource monitors and consumption monitoring across business domains.
- Contributed to platform run activities: service levels, monitoring and alerting, incident handling and lifecycle management of platform services.

Governance, security & compliance

- Implemented access and security controls across enterprise data assets: RBAC, Row-Level Security, GDPR compliance, traceability and auditability, in line with banking regulatory requirements including DORA.
- Supported data quality and governance practices in collaboration with the Data Office.

Leadership

- Coordinated Data, Engineering, Architecture, AI, DevOps and business stakeholders throughout the migration and the platform evolution.
- Drove the shift from individual use cases towards reusable and self-service Data & AI platform capabilities.

TECHNICAL ENVIRONMENT

Snowflake ,Power BI (semantic models, Power Query, DAX, RLS, workspaces, deployment pipelines) ,Dataiku ,Databricks , MLflow ,Python, SQL, Spark, Kafka , Azure , Docker, GitLab CI/CD ,OpenAI, Azure OpenAI, Claude, Mistral, LLaMA, LangChain,FAISS, PGVector, hybrid search, reranking, RAGAS ,Azure Document Intelligence, VLMs, Docling FastAPI, REST APIs, Redis Streams , RBAC, RLS, GDPR, DORA, encryption, auditability

DATA PLATFORM LEAD / LEAD DATA & AI PLATFORM : TotalEnergies

12/2020 - 12/2025

Design, industrialisation and evolution of an enterprise Data Platform on Snowflake serving multiple business domains (Retail, Human Resources, industrial operations), with a strong focus on Master Data Management, data quality, governance, BI and self-service analytics — and providing the data foundation for Machine Learning and Generative AI use cases.

KEY ACHIEVEMENTS

Platform & architecture

- Contributed to the design and evolution of the enterprise Data Platform, supporting multiple business domains and large volumes of heterogeneous data.
- Designed and managed data flows integrating data from multiple enterprise applications and operational systems into Snowflake.
- Worked on Snowflake data architecture, data modelling and trusted data layers supporting analytics, reporting, operational processes and AI use cases.
- Designed and industrialised scalable data pipelines with Databricks, PySpark, dbt, Airflow and Kafka, covering both batch and streaming ingestion.
- Defined data architecture and technical standards covering data modelling, MDM, APIs, governance, security and platform integration.

Data products, BI & self-service

- Worked closely with Retail, HR and business stakeholders to translate data and reporting requirements into reusable data products and platform capabilities.
- Designed and maintained Power BI datasets, semantic models, dashboards and KPIs used by business teams for operational monitoring, performance analysis and decision-making.
- Enabled self-service analytics by providing governed and trusted datasets to business users.

Data management & governance

- Contributed to Master Data Management initiatives improving the consistency, quality, governance and reuse of key enterprise data across business domains.
- Implemented data quality, governance, lineage and access-control practices across enterprise data assets, including RBAC, GDPR and auditability.

AI industrialisation

- Supported the industrialisation of Machine Learning use cases using enterprise Data Platform capabilities, Databricks and MLflow.
- Built the data foundations required for Generative AI, including document ingestion, knowledge bases and RAG pipelines.
- Designed and deployed AI agents and Document AI solutions for information extraction, document analysis, summarisation and business process automation.
- Implemented MLOps / LLMOps practices covering CI/CD, monitoring, evaluation, observability and continuous improvement.

Leadership

- Coordinated multidisciplinary teams of approximately 20 people across business, Data Engineering, Data Science, Architecture, Cloud and DevOps.
- Managed external partners and service providers contributing to platform delivery.
- Supported the adoption and continuous evolution of data products, BI capabilities and AI services across the organisation.

TECHNICAL ENVIRONMENT

Snowflake, Databricks, dbt, Azure, Python, SQL, PySpark, Spark, Kafka, Airflow, Power BI (datasets, dashboards, KPIs, self-service analytics), MDM, data modelling, data quality, data governance, data lineage, MLflow, OpenAI, Claude, LLaMA, Mistral, LangChain, LlamaIndex, Azure Document Intelligence, Docling, Tesseract, OpenCV, FAISS, PGVector, BM25, hybrid search, reranking, FastAPI, REST APIs, Kafka, Redis Streams, PostgreSQL, MongoDB, Redis, Prometheus, Grafana, RBAC, GDPR, auditability

HEAD OF DATA / CHAPTER DATA LEAD : RATP, MSM programme (RATP, Alstom, Renault, IRT SystemX)

08/2018 - 12/2020

Data leadership within a large operational organisation and a multi-partner mobility programme, focused on building Data Platform capabilities, data products and business analytics, and on industrialising data-driven use cases for operational teams.

KEY ACHIEVEMENTS

- Defined and implemented Data Platform capabilities supporting operational reporting, analytics and Machine Learning use cases.
- Built and deployed a multi-layered Data Lake on Azure for scalable data ingestion, processing and analytics, with Kafka for secure streaming and real-time pipelines.
- Translated business requirements into data products, datasets, KPIs and reusable data services.
- Developed and maintained Power BI dashboards and reporting capabilities, translating operational requirements into actionable KPIs and business visualisations.
- Designed microservices-based architectures for real-time supervision and incident management.
- Developed predictive models (LSTM, Random Forest, K-Means) for passenger flow, incident duration and operational planning, and OCR / classification pipelines (Tesseract, OpenCV, RNN, CNN) for unstructured data processing.
- Established data quality, governance and access-control practices for reliable and secure business reporting, including anomaly detection and log analysis for regulatory compliance.
- Managed internal and external project teams (AI engineers, data scientists, developers): scope definition, resource planning, roadmap alignment, technical governance and delivery standards.
- Provided training to business and technical teams on data platforms, data visualisation and process automation.
- Delivered functional AI demonstrators for smart mobility supervision at La Défense, improving operational safety and monitoring.

TECHNICAL ENVIRONMENT

Azure (Synapse Analytics, IoT, Functions, OneLake / Delta Lake), GCP (BigQuery, Dataflow, Dataproc, Pub/Sub, AI Platform), Kafka, Dataiku, Talend, Power BI, Python (TensorFlow, NLTK, OpenCV), OCR, Docker, GitLab, MongoDB, PostgreSQL, ELK stack

Publication & references: Architecture of a Public Transport Supervision System Using Hybridization Models Based on Real and Predictive Data, IEEE / DSD 2020 (pp. 440-446), MSM project: irt-systemx.fr/en/projets/msm

LEAD DATA ENGINEER / DATA SCIENTIST — FINANCIAL CRIME & DATA GOVERNANCE — BPCE, Banking & Insurance

01/2017 - 08/2018

Data and AI programme within a major French banking and insurance group, covering anti-money-laundering (AML), counter-terrorism financing (CTF), sanctions and embargo screening, fraud detection and GDPR compliance of financial data processing. The mission combined the industrialisation of detection engines with the structuring of the group's data governance foundations.

KEY ACHIEVEMENTS

Data governance, catalog & lineage

- Designed and deployed the enterprise data dictionary (data catalog) and data lineage capability using Apache Atlas, establishing traceability of financial data assets from source systems through to detection engines and regulatory reporting.
- Structured metadata, ownership and data definitions across AML, sanctions and fraud domains, improving the discoverability and reuse of trusted data assets.
- Implemented data quality controls, anomaly detection and log analysis on financial data feeds, supporting auditability and regulatory evidence.

Regulatory compliance & risk data

- Ensured GDPR compliance of financial data processing chains, covering access control, traceability and data protection requirements.
- Delivered structured reports and Power BI monitoring dashboards for fraud analytics and risk teams, translating detection outcomes into risk KPIs used in steering committees.
- Defined key performance indicators for the detection systems (precision, false-positive rate, alert throughput) and tracked them as service-level metrics of the detection platform.
- Analysed fraud patterns and formulated preventive AI strategies presented to risk and compliance stakeholders.

Detection platform & AI industrialisation

- Built an end-to-end AI-driven anti-fraud engine for banking and insurance operations, and automated the alerts, sanctions and embargo management workflows through combined machine learning and rule-based engines.
- Reduced false positives significantly through NLP-based confusion-matrix analysis (TF-IDF, hashing vectorisation) and classification models triaging alerts into PASSED and FAILED categories.
- Developed a fuzzy and phonetic matching engine to identify high-risk links between suspect entities and insurance clients — a core screening capability for sanctions and watch-list monitoring.
- Deployed Elasticsearch / Kibana clusters for large-scale log indexing, alert investigation and fraud visualisation.
- Enabled real-time monitoring and optimised resource allocation across the detection pipeline through automated processes.

Delivery & engineering practices

- Applied DevOps and TDD practices with GitLab CI/CD for continuous integration, testing and deployment of models into production.
- Estimated defect-correction throughput and release reliability prior to production go-live.
- Ran Agile Scrum adapted to data-centric delivery in a banking environment: kickoff, steering committees, retrospectives, testing and go-live reviews.

TECHNICAL ENVIRONMENT

Python (scikit-learn, NLTK, H2O.ai), Spark, AWS, SQL, Apache Atlas (data catalog & lineage), Elasticsearch, Logstash, Kibana, Power BI, GitLab CI/CD, NLP, fuzzy and phonetic matching, AML / CTF, sanctions and embargo screening, GDPR, Agile Scrum, DevOps, TDD

LEAD DATA SCIENTIST — Luxury Retail

09/2015 - 01/2017

KEY ACHIEVEMENTS

- Built a customer intelligence platform: CRM and transactional data integration (Oracle SQL to Cassandra via Sqoop), Kafka streaming ingestion, and enrichment with external web-crawled datasets for customer segmentation.
- Defined and tracked retail performance indicators: sales by category, return rates, churn and retention, store visit frequency, and sales growth by item, store, department and country.
- Developed churn, behaviour prediction and recommendation models (collaborative filtering), and industrialised the model code for deployment.
- Monitored real-time KPIs (revenue, churn, customer satisfaction) through Spark, Storm and Power BI dashboards, and delivered insights to marketing and operations.

TECHNICAL ENVIRONMENT

Python (scikit-learn, NLTK, H2O), Hadoop, Spark, Storm, Kafka, Cassandra, Oracle SQL, Sqoop, ETL, Power BI

DATA SCIENTIST — Beamlabs / Française des Jeux (FDJ)

11/2014 - 09/2015

KEY ACHIEVEMENTS

- Built a Big Data foundation for real-time marketing: distributed Hadoop and HBase architecture, Kafka and ZooKeeper streaming ingestion, Talend ingestion into Elasticsearch.
- Migrated and merged relational databases (Oracle SQL) to NoSQL (Cassandra) using Sqoop, and enriched internal data through web scraping and distributed crawling.
- Developed a real-time recommendation engine and predictive marketing models based on player behaviour analysis, with hyperparameter tuning and cross-validation.

TECHNICAL ENVIRONMENT

Python (scikit-learn, NLTK, H2O), Spark, SQL, Hadoop, HBase, Hortonworks, Kafka, ZooKeeper, Elasticsearch, Kibana, Talend

R&D ENGINEER — PhD, CentraleSupélec / École Polytechnique

02/2010 - 10/2014

KEY ACHIEVEMENTS

- Applied machine learning to the optimisation of photovoltaic cell performance and to the modelling of graphene-based transistors; conducted a market analysis of photovoltaic expansion in Europe.
- Developed simulation and data analysis algorithms in Python and R (FEM, FDTD, crystallographic data analysis) and geographic / scientific data visualisation with R-Shiny.

TECHNICAL ENVIRONMENT

Python, R, R-Shiny, Lumerical, Java

EDUCATION

- PhD, CentraleSupélec — Université Paris-Saclay, Orsay (2011 - 2014)
- Master of Sciences, Université Pierre et Marie Curie — Paris VI (2010 - 2011)
- Maîtrise en Sciences de l'Ingénieur, Université Paris Diderot — Paris VII (2009 - 2010)

COMMUNICATIONS & PUBLICATIONS

- Architecture of a Public Transport Supervision System Using Hybridization Models Based on Real and Predictive Data — IEEE / DSD 2020, pp. 440-446
- Disruption Impact on the Offer and Demand on the Public Transport Network in Paris
- Big Data Paris 2018 — speaker: "We can tell you by what external context your sales are influenced"

INSTRUCTOR EXPERIENCE

- Advanced Python for Machine Learning — 5 days, SQLI Institute, 2019
- Datalab — 3 days, Saagie, 2019
- SQL for Data Science — 4 days, Fitec, 2018
- ELK — 5 days, Fitec, 2017
- Kafka — 5 days, Air4Data / SQLI Institute, 2017
- DataViz (Tableau vs Power BI) — 4 days, SQLI Institute, 2017